



Figure 4: OpenCV4Android development modes

- d. The application starts and runs.
3. For successive apps:
 - a. The application starts without any additional overhead.
 - b. The target device's OpenCV library updates will be suggested, if there are any.

OpenCV4Android: Development modes

OpenCV4Android runs two development modes, as shown in Figure 4. One of the modes is to completely use the OpenCV4Android Java API. As stated earlier, OpenCV is written in the C/C++ programming languages. The calls made using the Java API are translated using JNI (Java Native Interface), which has to do the wrapping at the beginning of the call. Once the execution is completed, the results are returned to the API and hence another wrapping is done at this stage.

The other mode is to use the native code itself (C++) in the Android application. Here, the Android native development kit (NDK) needs to be configured for accessing the native code. The native code runs faster than the Java API as it doesn't involve additional overhead. Nevertheless, the power of the Java API shouldn't be underestimated. For many of the routine tasks, performance is at accepted levels.

Setting up the OpenCV4Android SDK

The OpenCV4Android SDK is downloaded as a zip file. Extract the zip file into a folder like C:\OpenCV4Android. If you already have the Android development environment in your system, then create a new workspace. Import the projects available in the extracted folder by pointing at the root. At this stage, you will be able to see around eight entries in the list. Select all of them and import them into your workspace. The detailed set-up instructions are available at http://docs.opencv.org/doc/tutorials/introduction/android_binary_package/O4A_SDK.html.

Some of the projects in the list, like face detection, require native code support. Hence the Android NDK has to be configured to launch these applications. The Android NDK can be downloaded and configured from <https://developer.android.com/tools/sdk/ndk/index.html>. The detailed

configuration instructions are available at <https://web.stanford.edu/class/ee368/Android/Tutorial-2-OpenCV-for-Android-Setup-Windows-API11.pdf>.

Once the SDK is configured, learners can experiment with the computer vision algorithms through the sample programs provided.

The OpenCV Vision Challenge for developers and researchers

In November 2014, the OpenCV Foundation launched a challenge with support from DARPA and Intel Corporation. The aim was to promote community participation and improve the OpenCV library. The challenge invites developers and researchers to submit new computer vision algorithms or new implementations of existing algorithms. Rewards worth US\$ 50,000 await successful developers. The 11 computer vision areas for which the algorithms are required are listed below:

1. Image segmentation
2. Image registration
3. Human pose estimation
4. SLAM
5. Multi-view stereo matching
6. Object recognition
7. Face recognition
8. Gesture recognition
9. Action recognition
10. Text recognition
11. Tracking

The deadline for submissions is May 8, 2015, and the winners will be announced on June 8, 2015 at CVPR 2015. For more detailed information you can visit <http://code.opencv.org/projects/opencv/wiki/VisionChallenge> and <http://code.opencv.org/attachments/1672/OpenCVVisionChallenge.pdf> 

References

- [1] <https://web.stanford.edu/class/ee368/Android/Tutorial-2-OpenCV-for-Android-Setup-Windows-API11.pdf>
- [2] <http://opencv.org>
- [3] http://docs.opencv.org/doc/tutorials/introduction/android_binary_package/dev_with_OCV_on_Android.html#dev-with-ocv-on-android
- [4] http://docs.opencv.org/doc/tutorials/introduction/android_binary_package/dev_with_OCV_on_Android.html
- [5] <http://docs.opencv.org/platforms/android/service/doc/Intro.html>
- [6] <https://developer.android.com/tools/sdk/ndk/index.html>

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